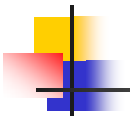


Programming Java

Networking

Incheon Paik

Contents

Internet Addresses

- Transmission Control Protocol (TCP) : obtain reliable, sequenced data exchange.
- User Datagram Protocol (UDP) : obtain a more efficient, best-effort delivery.

GetByName() Method

static InetAddress getByName (String hostName) throws UnknownHostException

getAllByName() Method

static InetAddress[] getAllByName (String hostName) throws UnknownHostException

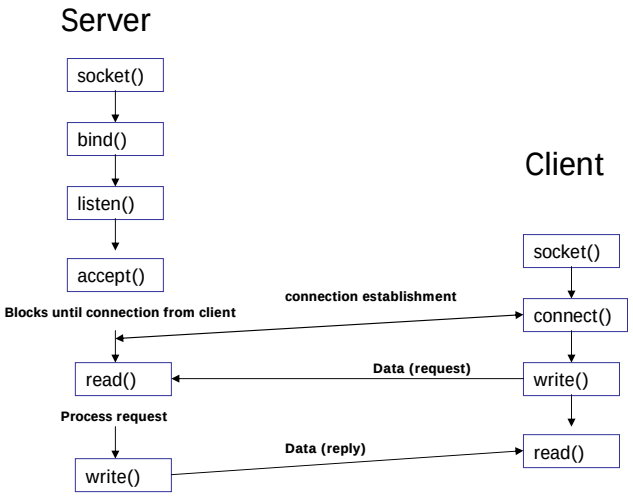
getLocalHost() Method

static InetAddress getLocalHost() throws UnknownHostException

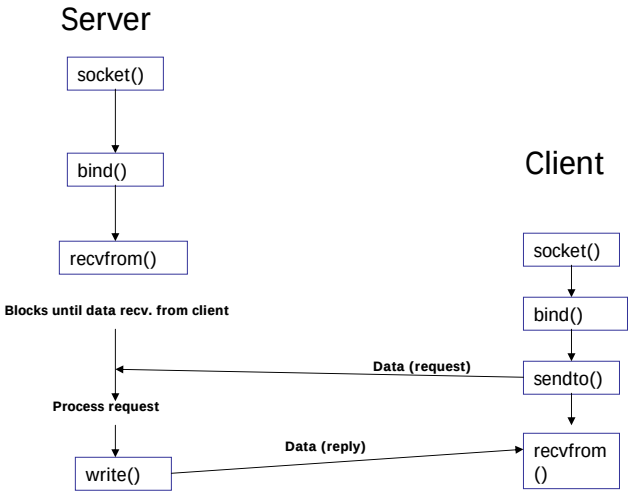
```
import java.net.*;

class InetAddressDemo {
  public static void main(String args[]) {
    try {
      InetAddress ias[] =
        InetAddress.getAllByName(args[0]);
      for (int i = 0; i < ias.length; i++) {
        System.out.println(ias[i].getHostName());
        System.out.println(ias[i].getHostAddress());
        byte bytes[] = ias[i].getAddress();
        for (int j = 0; j < bytes.length; j++) {
          if (j > 0)
            System.out.print(".");
          if (bytes[j] >= 0)
            System.out.print(bytes[j]);
          else
            System.out.print(bytes[j] + 256);
        }
        System.out.println("");
      }
    }
    catch (Exception e) {
      e.printStackTrace();
    }
  }
}
```

http://java.sun.com/j2se/1.5.0/docs/api/java/net/InetAddress.html

Socket Call for Connection-Oriented Protocol



Socket Call for Connectionless Protocol



Server Sockets and Sockets

ServerSocket Constructor

ServerSocket(int port) throws IOException

accept() Method

Socket accept() throws IOException

close() Method

void close() throws IOException

Socket Class

Socket(String hostName, int port) throws UnknownHostException, IOException

getInputStream(), getOutputStream Method

InputStream getInputStream() throws IOException
OutputStream getOutputStream() throws IOException

close()

void close() throws IOException

http://java.sun.com/j2se/1.5.0/docs/api/java/net/ServerSocket.html
http://java.sun.com/j2se/1.5.0/docs/api/java/net/Socket.html

Server Sockets and Sockets

```
import java.io.*;
import java.net.*;
import java.util.*;

class ServerSocketDemo {
  public static void main(String args[]) {
    try {

      // Get Port
      int port = Integer.parseInt(args[0]);
      Random random = new Random();
      //Create Server Socket
      ServerSocket ss = new ServerSocket(port);
      //Create Infinite Loop
      while(true) {
        //Accept Incoming Requests
        Socket s = ss.accept();

        //Write Result to Client
        OutputStream os = s.getOutputStream();
        DataOutputStream dos = new
DataOutputStream(os);
        dos.writeInt(random.nextInt());

        //Close socket
        s.close();
      }
    }
    catch (Exception e) {
      System.out.println("Exception: " + e);    }
  }
}
```

```
class SocketDemo {
  public static void main(String args[]) {
    try {
      //Get Server and Port
      String server = args[0];
      int port = Integer.parseInt(args[1]);
      //Create socket
      Socket s = new Socket(server, port);
      //Read random number from server
      InputStream is = s.getInputStream();
      DataInputStream dis = new
DataInputStream(is);
      int i = dis.readInt();
      //Display Result
      System.out.println(i);
      //Close Socket
      s.close();
    }
    catch (Exception e) {
      System.out.println("Exception: " + e);    }
  }
}
```

Running :

% java ServerSocketDemo 4321

% java SocketDemo 127.0.0.1 4321

Datagram Sockets and Packets

• UDP does not guarantee reliable, sequenced data exchange, and therefore requires much less overhead.

DatagramPacket Constructor

DatagramPacket(byte buffer[], int size)

DatagramPacket(byte buffer[], int size, InetAddress ia, int port)

DatagramSocket() Method

DatagramSocket() throws SocketException
DatagramSocket(int port) throws SocketException

receive() Method

void receive(DatagramPacket dp) throws IOException

send() Method

void send(DatagramPacket dp) throws IOException

close() Method

void close()

http://java.sun.com/j2se/1.5.0/docs/api/java/net/DatagramPacket.html

Datagram Sockets and Packets

```
class DatagramReceiver {
  private final static int BUFSIZE = 20;
  public static void main(String args[]) {
    try {
      //Obtain port
      int port = Integer.parseInt(args[0]);

      //Create a DatagramSocket object for the port
      DatagramSocket ds = new DatagramSocket(port);

      //Create a buffer to hold incoming data
      byte buffer[] = new byte[BUFSIZE];

      //Create infinite loop
      while(true) {
        //Create a datagram packet
        DatagramPacket dp =
          new DatagramPacket(buffer, buffer.length);

        //Receive data
        ds.receive(dp);

        //Get data from the datagram packet
        String str = new String(dp.getData());

        // Display the data
        System.out.println(str);
      }
    }
    catch (Exception e) {
      e.printStackTrace();
    }
  }
}
```

```
class DatagramSender {
  public static void main(String args[]) {
    try  {
      // Create destination Internet address
      InetAddress ia =
        InetAddress.getByName(args[0]);
      // Obtain destination port
      int port = Integer.parseInt(args[1]);

      // Create a datagram socket
      DatagramSocket ds = new DatagramSocket();

      //Create a datagram packet
      byte buffer[] = args[2].getBytes();
      DatagramPacket dp =
        new DatagramPacket(buffer, buffer.length,
          ia, port);
      // Send the datagram packet
      ds.send(dp);
    }
    catch (Exception e) {
      e.printStackTrace();
    }
  }
}
```

Running :

% java DatagramReceiver 4321

% java DatagramSender localhost 4321 Message

Uniform Resource Locators (URL)

URL

Protocol://host:/port/file

openStream() Method

InputStream() throws IOException

URL Constructor

URL(String protocol, String host, int port, String file) throws MalformedURLException

URL(String protocol, String host, String file) throws MalformedURLException

URL(String urlString) throws MalformedURLException

getFile(), getHost(), getPort(), and getProtocol() Methods

String getFile()

String getHost()

int getPort()

String getProtocol()

Refer http://java.sun.com/j2se/1.5.0/docs/api/java/net/URL.html

URL Demo Example

```
class URLDemo {
  public static void main(String args[]) {
    try {

      // Obtain URL
      URL url = new URL(args[0]);

      // Obtain input stream
      InputStream is = url.openStream();

      // Read and display data from URL
      byte buffer[] = new byte[1024];
      int i;
      while((i = is.read(buffer)) != -1) {
        System.out.write(buffer, 0, i);
      }
    }
    catch (Exception e) {
      e.printStackTrace();
    }
  }
}
```

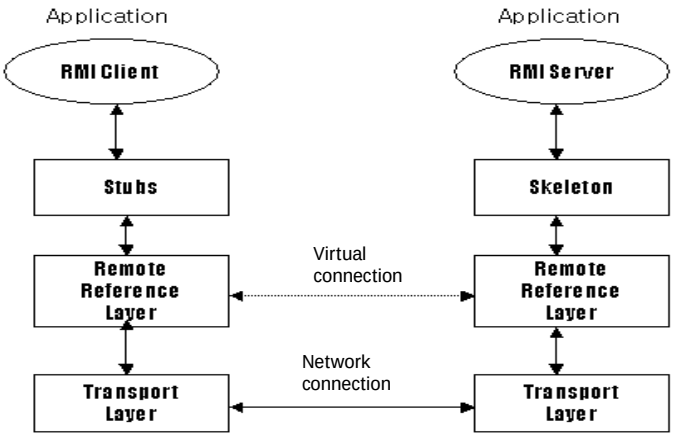
Run :

java URLDemo http://www.u-aizu.ac.jp

Introduction to RMI

- Distributed Processing on Network
- Define of Remote Interface
- Object Serialization
- java.rmi and java.rmi.server
- Create Stub and Skeleton

RMI Architecture



Example of RMI Implementation

- Interface Definition
- Implementation Class and Compilation
- Creation of Stub and Skeleton using rmic
- Creation of Server Application and Compilation
- RMI Registry and Start Server Program
- Creation of Client Program and Compilation
- Test Client

Hello Example : RMI

Interface

```
package examples.hello;

import java.rmi.Remote;
import java.rmi.RemoteException;

public interface Hello extends Remote {
    String sayHello() throws RemoteException;
}
```


Hello Example : RMI

Implementation

```
package examples.hello;

import java.rmi.Naming;
import java.rmi.RemoteException;
import java.rmi.RMISecurityManager;
import java.rmi.server.UnicastRemoteObject;

public class HelloImpl extends UnicastRemoteObject
    implements Hello {

    public HelloImpl() throws RemoteException {
        super();
    }

    public String sayHello() {
        return  "Hello World!";
    }

    public static void main(String args[]) {

        // Create and install a security manager
        if (System.getSecurityManager() == null) {
                    System.setSecurityManager(new
RMISecurityManager());
        }
```

```
try {
                    HelloImpl obj = new HelloImpl();
                    // Bind this object instance to the
name "HelloServer"
                    Naming.rebind("HelloServer", obj);
                    System.out.println("HelloServer bound
in registry");
        } catch (Exception e) {
                    System.out.println("HelloImpl err: " +
e.getMessage());
                    e.printStackTrace();
        }
    }
}
```

Compile & Skeleton Creation :

```
% javac examples/hello/Hello.java
% javac examples/hello/HelloImpl.java
% rmic examples.hello.HelloImpl
```

Hello Example : RMI

A Client Application

```
package examples.hello;

import java.rmi.Naming;
import java.rmi.RemoteException;

public class HelloClient {

  public static void main(String args[]) {
    String message = "Hello: This is my test message";

    // "obj" is the identifier that we'll use to refer
    // to the remote object that implements the "Hello"
    // interface
    Hello obj = null;

    try {
                    obj = (Hello)Naming.lookup("//" + "/HelloServer");
                    message = obj.sayHello();
                } catch (Exception e) {
                    System.out.println("HelloClient exception: " + e.getMessage());
                    e.printStackTrace();
                }
    System.out.println("Message = " + message);
  } // end of main
} // end of HelloClient
```


Hello Example : RMI

File “policy”

```
grant {
                // Allow everything for now
                permission java.security.AllPermission;
};
```

Start Registry Server & Run Server and Client

```
% rmiregistry &
% java –Djava.security.policy=policy examples.hello.HelloImpl
% javac examples/hello/HelloClient.java
% java [–Djava.security.policy=policy] examples.hello.HelloClient
```

Please ensure there is the “policy” file in the current directory

Exercise

- **Step 1, 2 (A BMI Server and a Simple Chatting Program)**
  Use the ServerSoket and Soket class
  Slide # 4-5
- **Step 3 (RMI Example)**
  Refer Slice #10-17